

Newman-Penrose Formalism Calculations

Stephani et al., Equation (22.70)

Metric

$$ds^2 = -e^{(-k\rho+kt)} dt \otimes dt + e^{(-k\rho+kt)} d\rho \otimes d\rho + dz \otimes dz + \rho^2 d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, \rho, z, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} e^{(-\frac{1}{2} k\rho + \frac{1}{2} kt)} \right) dt + \left(-\frac{1}{2} \sqrt{2} e^{(-\frac{1}{2} k\rho + \frac{1}{2} kt)} \right) d\rho$$

$$n = \left(\frac{1}{2} \sqrt{2} e^{(-\frac{1}{2} k\rho + \frac{1}{2} kt)} \right) dt + \left(\frac{1}{2} \sqrt{2} e^{(-\frac{1}{2} k\rho + \frac{1}{2} kt)} \right) d\rho$$

$$m = \left(\frac{1}{2} \sqrt{2} \right) dz + \left(\frac{1}{2} i \sqrt{2} \rho \right) d\phi$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \right) dz + \left(-\frac{1}{2} i \sqrt{2} \rho \right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} e^{(\frac{1}{2} k\rho - \frac{1}{2} kt)} \right) dt + \left(-\frac{1}{2} \sqrt{2} e^{(\frac{1}{2} k\rho - \frac{1}{2} kt)} \right) d\rho$$

$$n = \left(-\frac{1}{2} \sqrt{2} e^{\left(\frac{1}{2} k \rho - \frac{1}{2} k t \right)} \right) dt + \left(\frac{1}{2} \sqrt{2} e^{\left(\frac{1}{2} k \rho - \frac{1}{2} k t \right)} \right) d\rho$$

$$m = \left(\frac{1}{2} \sqrt{2} \right) dz + \left(\frac{i \sqrt{2}}{2 \rho} \right) d\phi$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \right) dz + \left(-\frac{i \sqrt{2}}{2 \rho} \right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} e^{\left(\frac{1}{2} k \rho - \frac{1}{2} k t \right)} \partial_t - \frac{1}{2} \sqrt{2} e^{\left(\frac{1}{2} k \rho - \frac{1}{2} k t \right)} \partial_\rho$$

$$\Delta = -\frac{1}{2} \sqrt{2} e^{\left(\frac{1}{2} k \rho - \frac{1}{2} k t \right)} \partial_t + \frac{1}{2} \sqrt{2} e^{\left(\frac{1}{2} k \rho - \frac{1}{2} k t \right)} \partial_\rho$$

$$\delta = \frac{1}{2} \sqrt{2} \partial_z + \frac{i \sqrt{2}}{2 \rho} \partial_\phi$$

$$\bar{\delta} = \frac{1}{2} \sqrt{2} \partial_z - \frac{i \sqrt{2}}{2 \rho} \partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2} e^{\left(\frac{1}{2} k \rho - \frac{1}{2} k t \right)}}{4 \rho}$$

$$\sigma = -\frac{\sqrt{2} e^{\left(\frac{1}{2} k \rho - \frac{1}{2} k t \right)}}{4 \rho}$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}e^{\left(\frac{1}{2}k\rho-\frac{1}{2}kt\right)}}{4\rho}$$

$$\lambda = -\frac{\sqrt{2}e^{\left(\frac{1}{2}k\rho-\frac{1}{2}kt\right)}}{4\rho}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = \frac{1}{4}\sqrt{2}ke^{\left(\frac{1}{2}k\rho-\frac{1}{2}kt\right)}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{ke^{(k\rho-kt)}}{2\rho}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{ke^{(k\rho-kt)}}{2\rho}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "N"